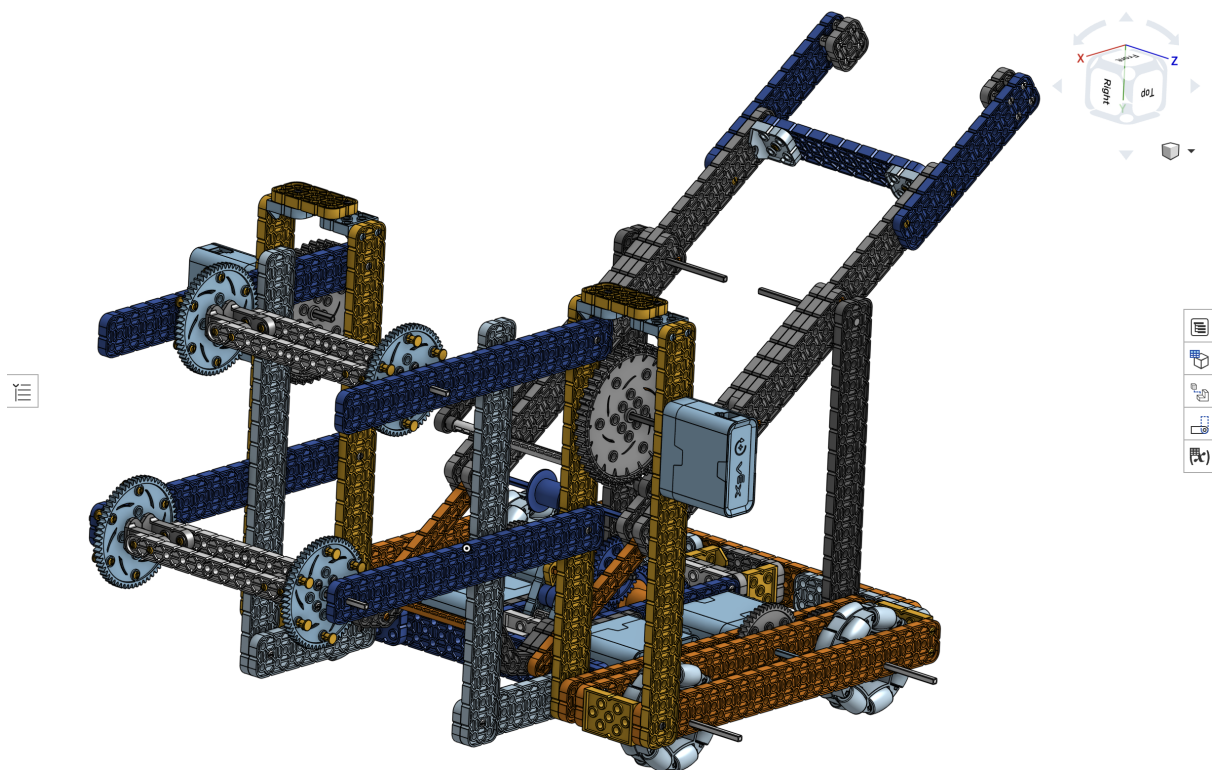


Lesson 20: 3D Modeling a Competition Robot for Rapid Relay



In this lesson, you'll create a detailed 3D model of a competition robot designed for the Rapid Relay challenge. Drawing on your knowledge of mechanical systems and 3D modeling techniques, you'll envision a robot capable of competing in the Rapid Relay competition. This is your opportunity to apply creativity and engineering principles to design a robot with both functionality and strategy in mind!

Instructions:

- Begin by reviewing the rules and objectives of the Rapid Relay challenge. Consider what kind of tasks your robot will need to perform, such as maneuvering through obstacles, interacting with game elements, or transporting objects.
- Sketch a rough draft of your robot design, identifying key features like the drivetrain, gear systems, lifting mechanisms, and any sensors you plan to

use.

- Using Onshape or your preferred 3D design software, create a 3D model of your robot based on your sketch. Pay close attention to dimensions, gear ratios, and the positioning of components to ensure your robot will function as intended.
- Include specific mechanisms that will allow your robot to excel in Rapid Relay, such as a reliable drivetrain for quick movement or a claw/lift system for object manipulation.
- Once your initial model is complete, refine the design for stability, durability, and efficiency. Think about the weight distribution, ease of assembly, and potential maintenance during the competition.
- Building instruction:

Lesson 20: Step-by-step building instruction for a Competition robot

By the end of this session, you'll have a complete 3D model of a competition-ready robot for the Rapid Relay event! This exercise will help you translate conceptual designs into detailed plans and prepare you for the next step: building your competition robot with VEX IQ parts.